

Corporate Governance, Financing Choices, and Visible Investment Signals in Indian FinTech Firms: Evidence from Public Data

Independent Research Project | Corporate Finance & Governance
Hand-Collected Panel Data • 25 Listed Indian FinTech-Linked Firms
• FY2020–21 to FY2023–24

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Abstract

Indian FinTech and technology-enabled financial services firms have expanded rapidly over the past decade, yet systematic, data-grounded evidence on how their financing choices and governance structures relate to observable investment behaviour remains limited. This study examines the association between corporate governance characteristics, financing choices and visible investment signals in a hand-collected panel of 25 Indian listed FinTech-linked firms across seven broad categories — digital payments, online insurance intermediation, FinTech infrastructure, core banking/BFSI software, digital broking/wealth-tech, and technology-enabled NBFC/digital lending — covering the financial years FY2020-21 to FY2023-24, yielding 100 firm-year observations. Data were extracted from four years of company annual reports and Corporate Governance Reports for each firm, in line with SEBI LODR disclosure requirements. Five variables were constructed: Investment (capital expenditure scaled by total assets), Leverage (total debt scaled by total assets), Return on Assets (net income scaled by total assets), Firm Size (natural logarithm of total assets), and Board Independence (independent directors as a proportion of board size). The study relies exclusively on descriptive statistics, year-on-year trend analysis, and pairwise Pearson correlation analysis; no regression, significance testing, or causal inference is undertaken.

The sample shows a mean investment ratio of 1.66% of assets (median 0.82%), a mean leverage ratio of 21.3% (median 7.1%), and a mean board independence of 52.96%. The correlation matrix shows that leverage is negatively associated with investment ($r = -0.380$), firm size is negatively associated with investment ($r = -0.277$), and board independence shows only a weak positive association with investment ($r = 0.078$). Return on assets is negatively associated with both firm size ($r = -0.534$) and board independence ($r = -0.408$). Year-on-year trend analysis shows that mean firm size and mean board independence both rose steadily across the sample period, while mean investment and mean leverage moved within a narrow range without a clear directional trend. These patterns are consistent with more indebted and larger firms in the sample exhibiting comparatively lower visible capital-expenditure intensity, while governance composition shows only a weak observed association with investment in this dataset. The principal limitation is the small, purposively selected sample (25 firms, 4 years) drawn from publicly available disclosures, which restricts statistical generalisability and precludes any causal interpretation of the reported associations.

Keywords: Corporate Governance, Financing Choices, Corporate Investment, Corporate Finance, Indian Technology Firms

1. Introduction and Research Questions

1.1 Background

The Indian FinTech and technology-enabled financial services sector has grown substantially since the mid-2010s, driven by digital payments adoption, the extension of listed market infrastructure institutions, the rise of digital broking and wealth-tech platforms, and the increasing use of technology in lending and core banking software. A number of firms in this space have listed on the NSE and/or BSE over the FY2017 to FY2023 window, which brings them under SEBI's Listing Obligations and Disclosure Requirements (LODR), requiring standardised disclosure of financial statements and Corporate Governance Reports, including board composition. This creates a public-data environment in which financing choices, governance characteristics and investment behaviour can, in principle, be observed and compared across firms and over time.

Financing decisions — how much of a firm's asset base is funded through debt relative to equity — are a central concern of corporate finance because they affect a firm's flexibility to fund future capital expenditure. Corporate governance mechanisms, particularly board composition and the proportion of independent directors, are widely regarded in the literature as monitoring devices intended to align management's investment decisions with shareholder interests. Corporate investment, commonly proxied by capital expenditure relative to total assets, is the visible outcome of these financing and governance conditions and is therefore a natural point at which to examine how the two might jointly relate to firm behaviour.

1.2 Research Motivation

Financing choices and governance characteristics are usually studied separately in isolated strands of the corporate finance literature. Examining them together, alongside a directly observable investment measure, offers a more integrated descriptive picture of how Indian FinTech-linked firms allocate capital. Because this sector combines firms of very different age, ownership history, and balance-sheet structure — from recently listed, loss-making digital platforms to long-established, debt-free software exporters — it provides a useful, if small, setting in which to describe how financing and governance profiles co-vary with visible investment signals, using only the firm's own publicly disclosed numbers.

1.3 Research Questions

- RQ1: How are financing choices associated with corporate investment in the sampled Indian FinTech-linked firms?
- RQ2: How are governance characteristics, specifically board independence, associated with investment decisions in the sample?
- RQ3: What patterns emerge when governance characteristics, financing choices and investment decisions are examined together across the sample and over the four-year period?

RQ3 is addressed descriptively, by considering the correlation matrix and the year-on-year trends jointly. The dataset does not contain an interaction term, a multivariate regression, or any moderation model; consequently, RQ3 does not test whether governance statistically moderates the financing–investment relationship. It instead reports the pattern of pairwise associations observed among the three sets of variables in the same sample.

2. Theoretical Background and Conceptual Framework

2.1 Financing Choices and Corporate Investment

Corporate finance theory has long linked capital structure to investment behaviour. Under the trade-off and pecking-order perspectives, higher leverage can constrain a firm's capacity to fund further capital expenditure, both because debt-servicing obligations absorb internally generated cash flow and because highly levered firms may face tighter external financing conditions. Financing-constraints research similarly suggests that firms with limited access to low-cost external capital — often smaller or more indebted firms — tend to show more conservative investment behaviour, while firms with stronger balance sheets and internally generated cash can sustain higher capital expenditure intensity relative to their asset base. These perspectives motivate an examination of how leverage, measured as total debt scaled by total assets, is associated with investment, measured as capital expenditure scaled by total assets, in the sampled firms.

2.2 Corporate Governance and Investment

A separate strand of literature examines the monitoring role of the board of directors, and particularly the proportion of independent directors, in shaping corporate investment decisions. Agency theory suggests that independent directors, being less connected to management, can help curb both under-investment arising from managerial risk aversion and over-investment arising from empire-building incentives, thereby improving the alignment between investment decisions and firm value. Board independence is accordingly treated in this study as an indicative governance-monitoring characteristic, measured as the number of independent directors divided by total board size, and is examined for its association with the investment ratio.

2.3 Conceptual Framework

Technology-enabled financial services firms present a distinctive combination of characteristics relevant to this framework: several are relatively young and recently listed, several report low or zero formal debt, and several have experienced substantial changes in board composition around their listing events. This combination of youth, variable capital structure and evolving governance makes the sector a relevant, if idiosyncratic, setting for observing how financing and governance characteristics co-vary with a visible investment signal. The conceptual logic guiding this study can be summarised as:

Corporate Governance → Financing Choices → Corporate Investment

Firm-level characteristics, specifically firm size (the natural logarithm of total assets) and profitability (return on assets), are also considered where available in the dataset, since both are commonly discussed alongside governance and financing variables in the corporate-finance literature. It must be stated clearly that this is a conceptual organising framework for presenting the analysis, not a causal or statistically tested model: the correlations reported later in this study describe association only, and no claim of directional causality between governance, financing and investment can be drawn from them.

3. Data, Sample and Variables

3.1 Sample Composition

The dataset comprises 25 Indian listed firms selected through purposive (judgmental) sampling, an approach the workbook's Sample Selection Criteria sheet identifies as appropriate for exploratory, panel-based governance-finance research on a still-emerging FinTech population in India. Firms were required to be listed on the NSE and/or BSE, to have a primary business model within one of the FinTech-linked categories described below, to have at least three consecutive years of publicly available financial statements and Corporate Governance Reports, and to have remained continuously listed and actively traded throughout the study period. The final sample spans FY2020-21 to FY2023-24 (four financial years), aligned to each firm's listing date; firms listed after the start of the period contribute a correspondingly shorter but still fully reported panel. With 25 firms observed over 4 years, the dataset contains 100 firm-year observations in the

Calculated Variables sheet, matching the descriptive-statistics sheet's reported sample size (N = 100 for Investment, Leverage, ROA and Firm Size; N = 97 for Board Independence).

As recorded in the Raw Data sheet, the 25 firms are distributed across the following business categories:

Category (as recorded in workbook)	Number of Firms
FinTech – Digital Payments	6
Digital Broking / Wealth-Tech	5
FinTech Infrastructure (RTA / Depository / Exchange)	4
FinTech – Core Banking / BFSI Software	4
Tech-enabled NBFC / Digital Lending	4
FinTech – Online Insurance Intermediation	1
Financial Services / Stock Broking	1
Total	25

Table: Sample composition by firm category (Raw Data sheet).

Three firm-years lack a board-composition figure — One97 Communications (Paytm) FY2021, PB Fintech FY2021, and Fino Payments Bank FY2021 — because these firms had not yet completed their public listing at the relevant fiscal year-end and pre-listing board disclosures were unavailable; these observations are recorded as missing (NA) rather than zero, as noted in the workbook.

3.2 Data Sources

All raw financial and governance figures (Assets, Debt, Equity, CAPEX, Revenue, Net Income, Cash, Board Size, Independent Directors) were hand-extracted from each firm's own four annual reports, as documented in the Data Source and Notes columns of the Raw Data sheet, with figures standardised to ₹ crore. Where firms reported zero debt, or negative shareholders' equity, these values were retained as reported rather than treated as missing, since the workbook's sample-selection notes state that they represent the firm's actual financial position.

3.3 Variables

The Calculated Variables sheet derives five variables from the raw data using live formulas, summarised below:

Variable	Measurement	Category / Purpose
Investment	Capital Expenditure ÷ Total Assets	Visible investment signal (dependent variable of interest)
Leverage	Total Debt ÷ Total Assets	Financing choice
ROA	Net Income ÷ Total Assets	Firm-level profitability
Firm Size	LN(Total Assets)	Firm-level characteristic
Board Independence	Independent Directors ÷ Board Size	Governance characteristic

Table: Variable definitions (Calculated Variables sheet).

No separate line item for capital expenditure exists in the raw data; the workbook uses the CAPEX column (purchase of property, plant and equipment plus intangible assets, from each firm's cash-flow statement) as the investment numerator. Where Debt was reported as "Nil" it is treated as zero, not missing. Only the variables listed above — Investment, Leverage, ROA, Firm Size and Board Independence — are analysed in this study; no other calculated variable exists in the workbook.

4. Methodology

Consistent with the exploratory, descriptive purpose of this study, three analytical techniques are used: descriptive statistics, year-on-year trend analysis, and pairwise Pearson correlation analysis. No regression model, interaction term, or statistical significance test is estimated, and no causal claim is made from any of the reported results.

4.1 Descriptive Statistics

The workbook's Descriptive Statistics sheet reports the number of observations (N), mean, median, minimum, maximum and sample standard deviation (STDEV.S) for each of the five variables, calculated across all 100 firm-year observations in the Calculated Variables sheet. A separate baseline-year table reports the same statistics restricted to FY2020-21 (the first year of the panel, N = 25 firms), allowing the starting position of the sample to be distinguished from its overall four-year profile.

4.2 Trend Analysis

Year-on-year patterns in Investment, Leverage, ROA, Firm Size and Board Independence are examined by computing the mean of each variable across all sampled firms for each of the four fiscal years (FY2021 to FY2024), using the firm-year values recorded in the Calculated Variables sheet. This shows how the sample's average financing, governance and investment profile evolved over the observation window, without attributing any single year's movement to a specific firm or event unless independently noted in the underlying data.

4.3 Correlation Analysis

The workbook's Correlation Matrix sheet reports pairwise Pearson correlation coefficients among the five calculated variables, computed across the full panel of firm-year observations. A Pearson coefficient close to +1 or -1 indicates a strong linear association, values near 0.3 to 0.5 (in absolute terms) are treated as moderate, and values below roughly 0.2 are treated as weak, following common descriptive conventions; these thresholds are used only to describe the strength of association observed in this specific dataset and are not statistical significance thresholds. Every correlation coefficient reported in Section 5 is reproduced exactly as it appears in the workbook. Consistent with the brief for this study, a correlation is described throughout as an association — for example, “leverage is negatively associated with investment in the sample” — and never as evidence that one variable causes a change in another.

5. Results

5.1 Descriptive Statistics

Variable	N	Mean	Median	Minimum	Maximum	Std. Dev.
Investment	100	0.0166	0.0082	0.0004	0.1411	0.0207
Leverage	100	0.2131	0.0709	0.0000	0.7659	0.2762
ROA	100	0.0803	0.0531	-0.1859	0.4525	0.1074
Firm Size	100	8.1478	8.1738	4.0872	10.9711	1.2746
Board Independence	97	0.5296	0.5556	0.0000	0.7778	0.1360

Table 1: Overall descriptive statistics, FY2020-21 to FY2023-24, all firm-year observations (Descriptive Statistics sheet).

The average firm in the sample invests 1.66% of its total assets in capital expenditure in a given year, though the median (0.82%) is well below the mean, indicating that investment is right-skewed: a small number of firm-years with unusually high capital expenditure, such as CDSL's FY2023 ratio of 14.11%, pull the mean above the typical observation. Leverage shows a similar pattern — a mean of 21.3% but a median of only 7.1% — reflecting that many sampled firms (particularly software exporters and market-infrastructure firms such as Oracle Financial Services Software, Nucleus Software, CAMS, CDSL and BSE) carry little or no formal debt, while a smaller set of NBFC and lending-oriented firms carry substantially higher leverage. Board independence averages 52.96% across the 97 available firm-year observations. The minimum board-independence value of zero reflects Zaggie Prepaid Ocean Services Ltd. during its pre-listing years (FY2021–FY2023), when the available disclosures record a two-member board with no directors classified as independent. Following its September 2023 listing, the board expanded to seven directors, including four independent directors, raising board independence to 57.14% in FY2024.

The baseline-year (FY2020-21) statistics show a lower mean firm size (7.7518 versus 8.1478 overall) and a lower mean board independence (49.59% versus 52.96% overall), consistent with the sample's more recently listed firms entering the panel with smaller asset bases and less mature governance disclosures at the start of the observation window.

5.2 Trend Analysis

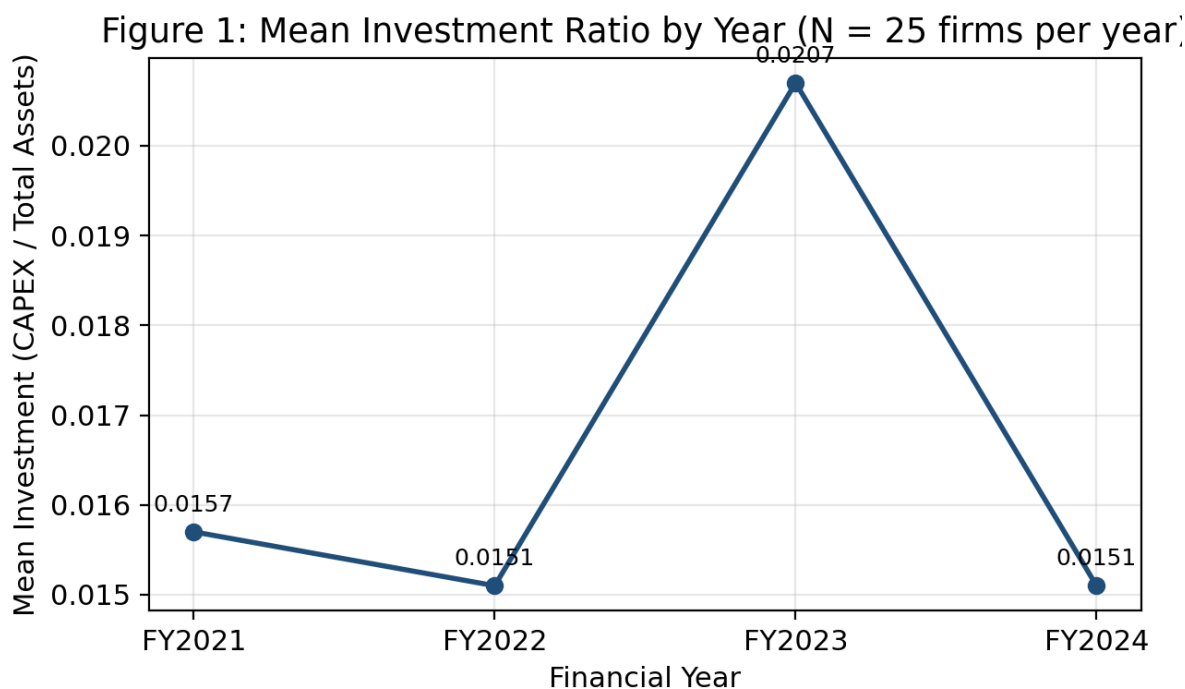


Figure 1: Mean investment ratio across the sample, FY2021–FY2024.

Year	Mean Investment	Mean Leverage	Mean ROA	Mean Firm Size	Mean Board Indep.
FY2021	0.0157	0.2155	0.0675	7.7518	0.4959
FY2022	0.0151	0.2186	0.0861	8.0781	0.5239
FY2023	0.0207	0.2099	0.0780	8.2470	0.5446
FY2024	0.0151	0.2086	0.0896	8.5143	0.5498

Table: Yearly mean values of the five calculated variables (computed from the firm-year panel in the Calculated Variables sheet).

Mean investment fluctuates within a narrow band across the four years (1.51% to 2.07% of assets), with a visible peak in FY2023 followed by a return to the FY2021–FY2022 level in FY2024, rather than following a steady upward or downward trend. Mean leverage is comparatively stable, declining only marginally from 21.55% in FY2021 to 20.86% in FY2024. In contrast, mean firm size and mean board independence both rise steadily and monotonically across all four years — firm size from 7.75 to 8.51 (log assets), and board independence from 49.59% to 54.98% — consistent with the sample's mix of firms growing in asset base and, on average, expanding independent representation on their boards over the period, alongside newly listed firms

completing their board disclosures. Mean ROA also rises from 6.75% in FY2021 to 8.96% in FY2024, with a temporary peak in FY2022 (8.61%), although the pattern was not monotonic.

5.3 Correlation Matrix

	Investment	Leverage	ROA	Firm Size	Board Indep.
Investment	1.0000	-0.3800	0.2089	-0.2774	0.0783
Leverage	-0.3800	1.0000	-0.1296	0.3063	-0.1700
ROA	0.2089	-0.1296	1.0000	-0.5337	-0.4081
Firm Size	-0.2774	0.3063	-0.5337	1.0000	0.4000
Board Indep.	0.0783	-0.1700	-0.4081	0.4000	1.0000

Table 2: Pairwise Pearson correlation matrix, calculated variables (Correlation Matrix sheet).

Figure 2: Leverage vs. Investment (Firm-Year Observations, N = 100)
Pearson $r = -0.380$

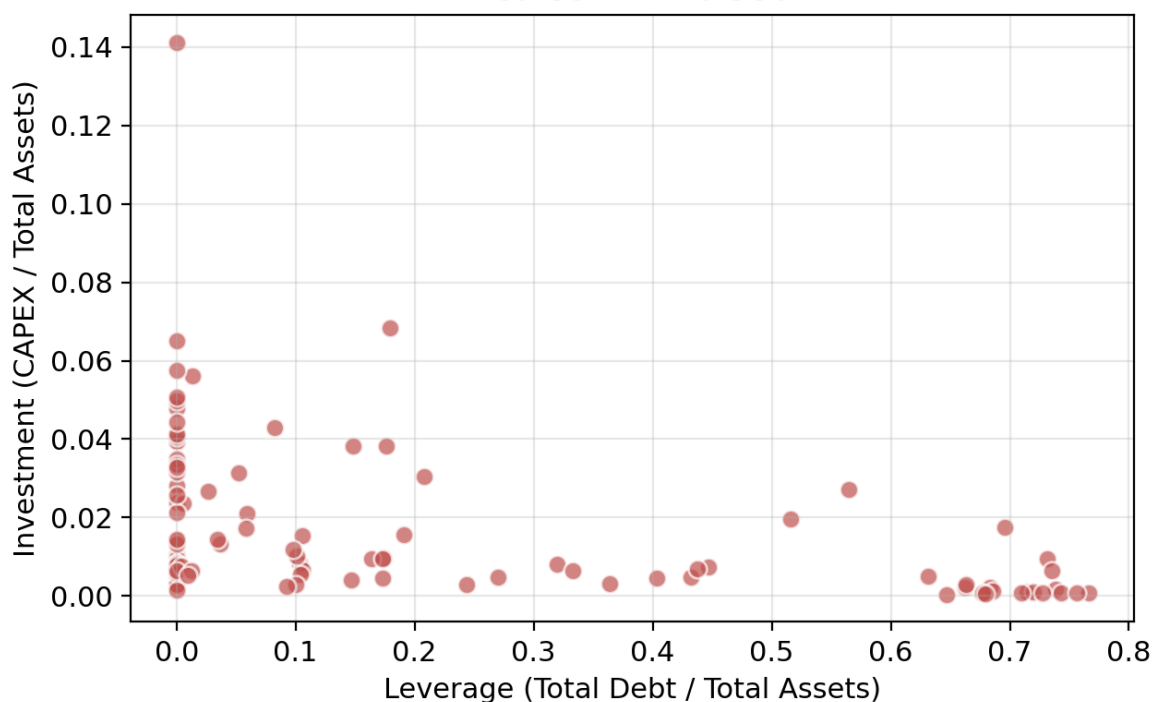


Figure 2: Leverage vs. Investment, firm-year observations (N = 100).

Figure 3: Board Independence vs. Investment (Firm-Year Observations, 1
Pearson $r = 0.078$

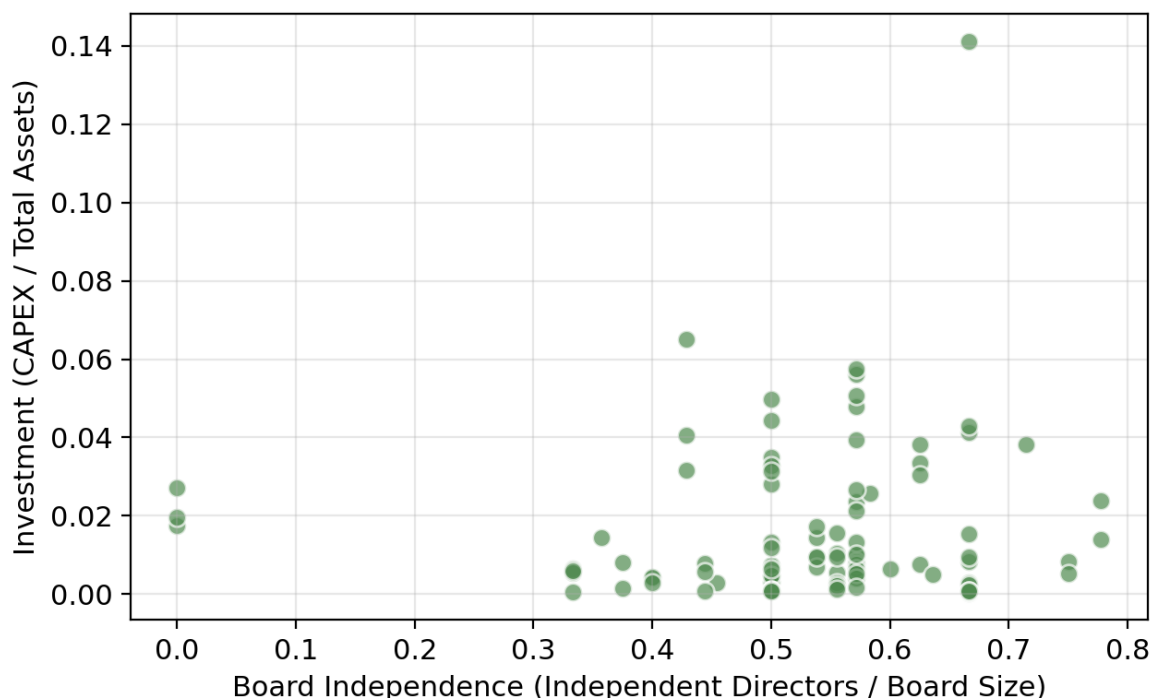


Figure 3: Board Independence vs. Investment, firm-year observations ($N = 97$).

The most economically meaningful association in the matrix is between leverage and investment ($r = -0.380$): firm-years with a higher debt-to-assets ratio in this sample tend to show a lower capital-expenditure-to-assets ratio, a moderate negative association visible in Figure 2, where firm-years above roughly 40% leverage (mostly NBFC/digital-lending and SBI Cards observations) cluster at low investment ratios below 1%. Firm size is also negatively associated with investment ($r = -0.277$): the largest firms in the sample by total assets, such as SBI Cards, Motilal Oswal and Credit Access Grameen, generally report lower investment ratios than smaller, growth-stage firms. Board independence shows only a weak positive association with investment ($r = 0.078$), visible in the dispersed pattern in Figure 3, indicating that in this sample, the proportion of independent directors is not strongly associated with the visible investment signal one way or the other. The strongest association in the entire matrix is between ROA and firm size ($r = -0.534$): larger firms in the sample tend to report lower profitability ratios, and ROA is also moderately negatively associated with board independence ($r = -0.408$). This pattern may reflect differences in listing maturity and governance development, although listing tenure is not directly tested in the present analysis.

6. Discussion

6.1 RQ1 — Financing and Investment

The negative association between leverage and investment ($r = -0.380$) is the clearest financing-related pattern in the data. Firms in the sample that carry a higher proportion of debt to total assets tend, on average, to show a lower capital-expenditure ratio in the same firm-year. What the data show is a moderate negative pairwise association; a possible economic interpretation, consistent with financing-constraints and debt-overhang reasoning discussed in Section 2.1, is that firms already carrying substantial debt may have less capacity or appetite to fund further visible capital expenditure. What cannot be concluded from this dataset is whether leverage actually restricts investment, whether lower-investment firms simply choose to finance themselves with more debt for unrelated reasons, or whether some third factor — such as business-model maturity — drives both variables jointly; the correlation coefficient alone cannot distinguish between these explanations.

6.2 RQ2 — Governance and Investment

Board independence shows only a weak positive association with investment ($r = 0.078$) in this sample. What the data show is that, across the 97 firm-years with disclosed board composition, the proportion of independent directors carries almost no linear relationship with the capital-expenditure ratio. A possible economic interpretation is that, within this particular sample of already SEBI-LODR-compliant listed firms — most of which report board independence at or above 40% — variation in board independence may be too narrow, or too disconnected from the specific capital-allocation decisions captured by CAPEX, to show a visible relationship with investment. What cannot be concluded is that board independence is unrelated to governance quality more broadly, or that it plays no monitoring role in investment decisions; the weak correlation reflects only the linear association observed in this particular sample and variable definition, not a test of the underlying governance-monitoring theory.

6.3 RQ3 — Governance, Financing and Investment Together

Considering the correlation matrix and the trend analysis together, two threads run through the data. First, firm size appears to connect governance and financing characteristics in this sample: larger firms show both higher leverage ($r = 0.306$) and higher board independence ($r = 0.400$), while simultaneously showing lower investment ($r = -0.277$) and lower profitability ($r = -0.534$). Second, the trend analysis shows that firm size and board independence both rose steadily across FY2021–FY2024, while investment and leverage stayed comparatively flat over the same period — suggesting that the sample's governance and scale characteristics evolved somewhat independently of its financing and visible investment behaviour. What the data show is a consistent pattern in which larger, more leveraged, more independently governed firms in this sample tend to report lower visible investment and lower ROA. A possible economic interpretation, linking to the agency-theory and capital-structure concepts discussed in Section 2, is that this reflects a maturity effect — larger, more established firms in the sample (several of

them long-listed software or market-infrastructure companies) may simply have passed their highest-growth, highest-CAPEX phase, while carrying more developed governance structures as a separate consequence of listing tenure. What cannot be concluded is that governance moderates the financing–investment relationship, since the dataset contains no interaction or multivariate model capable of testing that claim; RQ3 is answered here purely as a descriptive pattern across three separate pairwise associations observed in the same 25-firm, 4-year sample.

7. Limitations, Conclusion and Future Research

7.1 Limitations

- **Sample size:** the dataset comprises only 25 firms and 100 firm-year observations (97 for board independence), which limits the statistical generalisability of any reported association.
- **Observation period:** the four-year window (FY2020-21 to FY2023-24) is short and partly overlaps with the post-pandemic recovery period, which may itself have influenced profitability and investment patterns independent of financing or governance characteristics.
- **Reliance on publicly available information:** all figures were hand-extracted from published annual reports and Corporate Governance Reports; any inconsistency in how individual firms define or disclose CAPEX, debt or board composition (several of which are documented firm-by-firm in the workbook's Notes column) carries through into the calculated variables.
- **Measurement limitations of governance variables:** board independence, as defined here, captures only the proportion of independent directors and does not reflect other governance dimensions such as board diversity, audit-committee composition, promoter shareholding, or ownership concentration, none of which are present in this dataset.
- **Correlation rather than causal or econometric analysis:** the study reports descriptive statistics, trend patterns and pairwise Pearson correlations only; it does not control for confounding variables, does not test statistical significance, and cannot establish causality between financing, governance and investment.
- **Generalisability:** the purposively selected sample is anchored in six FinTech-linked categories among Indian listed firms and cannot be generalised to unlisted FinTech firms, non-Indian markets, or non-technology financial firms.

7.2 Conclusion

This study set out to describe how financing choices, governance characteristics and visible investment signals relate to one another in a hand-collected panel of 25 Indian listed FinTech-linked firms over FY2020-21 to FY2023-24. On RQ1, the data show a moderate negative association between leverage and investment ($r = -0.380$): more indebted firm-years in the sample tend to show lower capital-expenditure intensity. On RQ2, the data show only a weak

positive association between board independence and investment ($r = 0.078$), indicating no clear linear relationship between governance composition and the visible investment signal in this sample. On RQ3, the combined evidence points to firm size as the variable most consistently linked to both financing and governance characteristics: Larger firms in the sample are more leveraged and exhibit greater board independence, while also showing lower profitability and investment intensity. Over the sample period, mean firm size and board independence increased, whereas investment and leverage remained comparatively stable. For corporate finance and governance more broadly, the sample-level implication is that, among Indian FinTech-linked listed firms in this period, financing structure appears more closely tied to visible investment behaviour than does board composition, though both associations are best read as descriptive patterns within a single, small, purposively selected sample rather than generalisable findings.

7.3 Future Research

- Extend the sample to a **larger set of FinTech firms and a longer observation period** to test whether the observed patterns remain consistent over time.
- Incorporate **ownership and governance variables**, including multiple large shareholders, non-controlling block holders, promoter ownership, and audit-committee characteristics.
- Conduct **cross-country analysis across major emerging economies**, particularly India and China, to examine how different governance and regulatory environments shape firm behaviour.
- Extend the governance–finance framework to examine **corporate tax behaviour and risk-taking**, alongside financing and investment decisions.
- With a larger panel, apply **panel-data and multivariate methods** to formally test governance–financing–investment relationships and potential moderation effects.